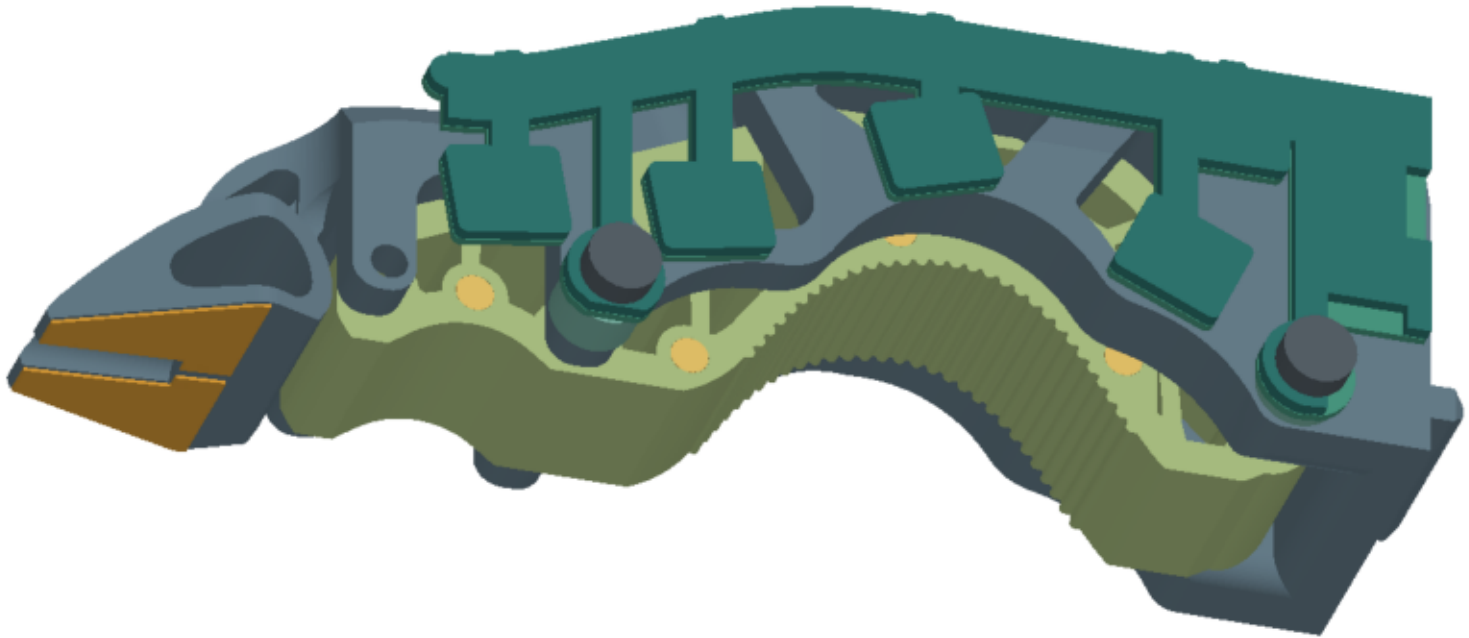
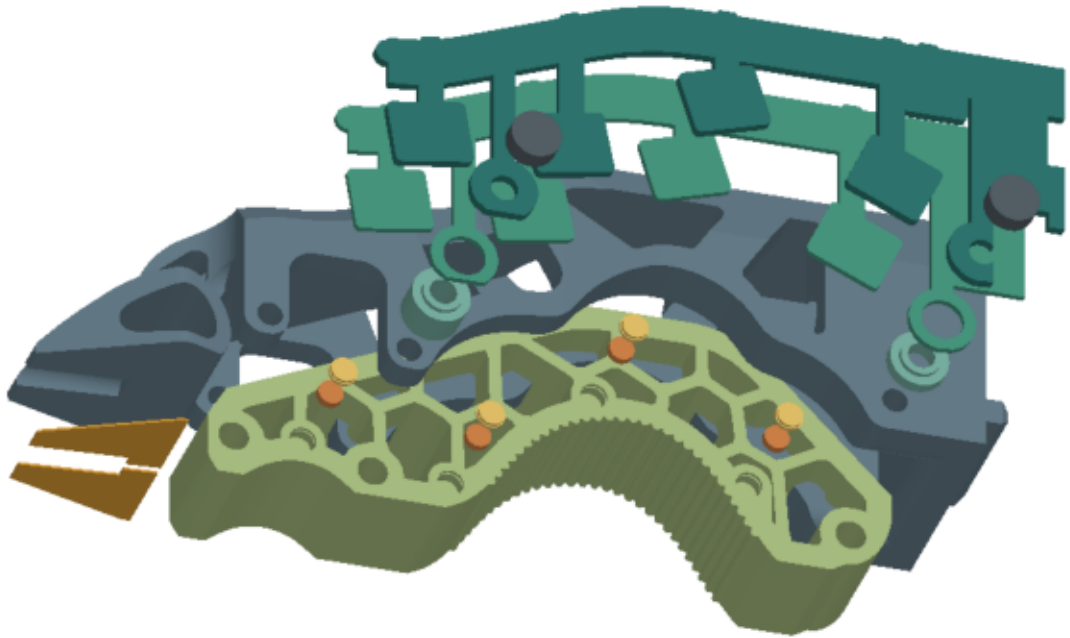
B2 · one PCB and removable backer



Original frame · B1 TPU/magnets · N0 nail pads · one PCB · two spacers · backer

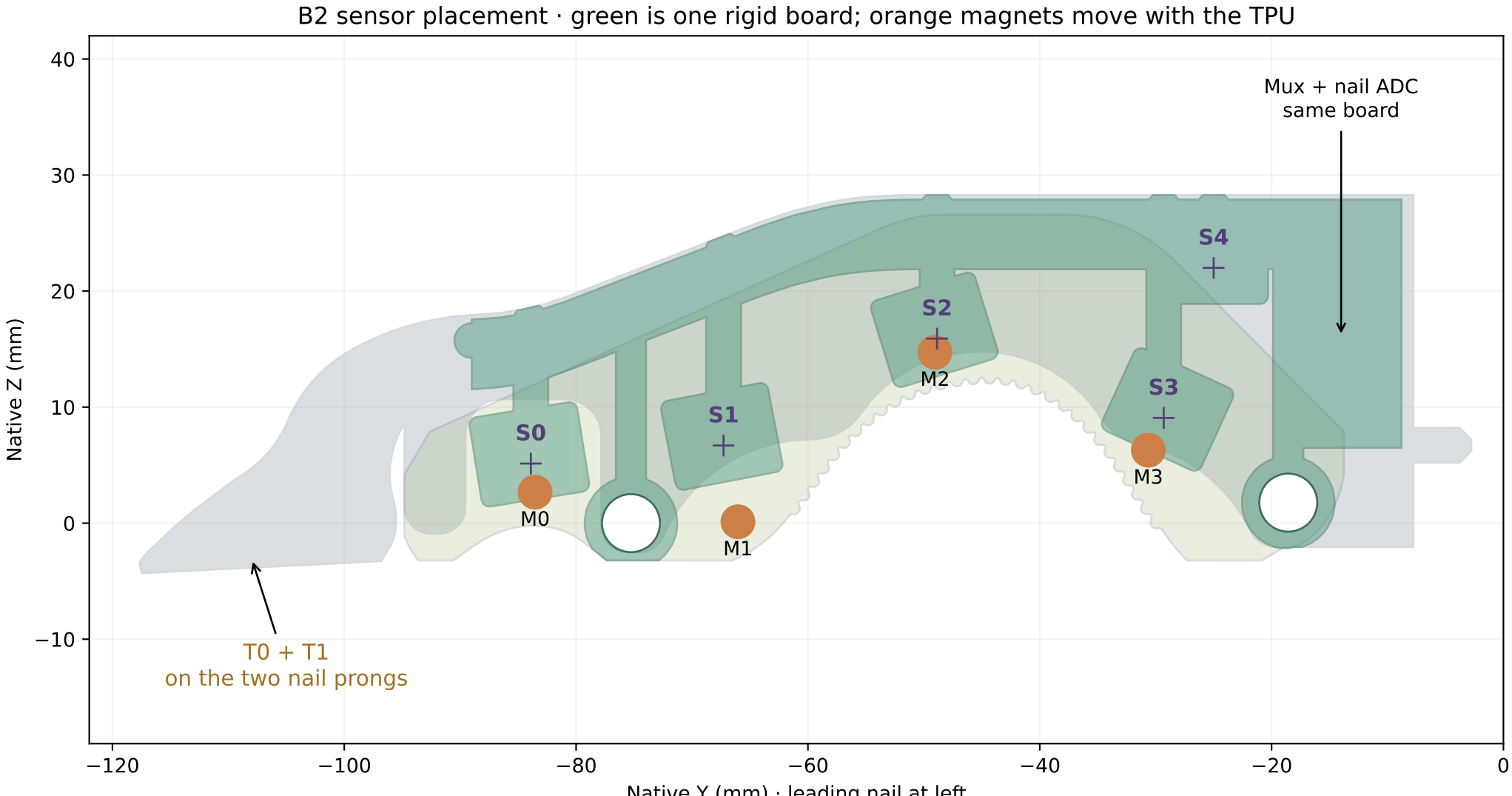
Actual CAD + component envelopes. Hardware unbuilt; cable joints and seating film are not pictured.

B2 · exploded prototype assembly



Original frame · B1 TPU/magnets · N0 nail pads · one PCB · two spacers · backer

Actual CAD + component envelopes. Hardware unbuilt; cable joints and seating film are not pictured.


B2 sensor placement · green is one rigid board; orange magnets move with the TPU
Native Z (mm)
Native Y (mm) · leading nail at left
40
30
20
10
0
−10
−120
−100
−80
−60
−40
−20
0
Mux + nail ADC
same board
S0
S1
S2
S3
S4
M0
M1
M2
M3
T0 + T1
on the two nail prongs

B2.1 manufacturing review · dimensions in mm · do not scale this page

Original ENPIRE hard frame, B1 TPU, magnets and sensor poses retained.
New hardware: one four-layer PCB, removable backer, two locating spacers.
PCB component plane X=8.4392; PCB thickness 0.80.
Backer outer X=10.5392; backing thickness 1.20; seating film 0.10.
Nominal minimum IC/body-to-frame gap: 0.450.
Added reach: 0.0.
Added global width: 3.650 before heads;
  6.050 with reserved heads; wires excluded.

PCB bores: 5.00. Spacer shoulder: 4.85 OD x 0.65 long; through bore: 3.20.
Spacer body: 7.00 OD. Keep PCB clamping annuli free of copper.
b2_locating_spacer_1: body length 4.4392; native Y/Z [-75.26417028511005, 0.0].
b2_locating_spacer_2: body length 1.5500; native Y/Z [-18.559447378460487, 1.771483509339382].

S0-S4: five MLX90393SLW-ABA-011 ICs; address 0x0C per isolated branch.
U10: TCA9548ARGER (RGE QFN-24), 0x70. U11: ADS1115IDGSR, 0x48.
J1: pin 1 3V3, 2 GND, 3 SCL/GP5, 4 SDA/GP4.
J2: pin 1 T0 excitation, 2 T0 sense, 3 T1 excitation, 4 T1 sense.
A 100k continuity resistor stays across each nail pad at the fingertip.

Print witnesses and measure installed fit before making a batch.
CAD and PCB checks do not establish force accuracy, 20 N capacity or lifetime.
Use the STEP masters, checked Gerbers/drills and complete B2 build guide.